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Q6 Problem Statement

The matrix of size $D \times D$ is defined as $A = xx^T + yy^T$, where $\vec{x}, \vec{y} \in \mathbb{R}^{D \times 1}$ are two non-zero vectors. Which of the following statements are true for this matrix.

1. One eigen-vector is y .
2. One eigen-vector is x .
3. One eigen-vector is z where $z \perp x$.
4. One eigen-vector is z where $z \perp x$ and $z \perp y$. ✓

Solution

Option one and 2 are unlikely, as $\vec{x}, \vec{y}$ should be orthogonal not one of them. Option 4 is likely to be true if $\vec{z}$ exists to be an eigen vector then it should be orthogonal to $\vec{y}$ and $\vec{x}$.

That is my best guess with vague analysis.

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Q7 Problem Statement

Let

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^{3 \times 1}$$

Further, Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Let $E = x^T A x$. Then, $\frac{\partial E}{\partial x}$ is:

1. ✓

$$\begin{bmatrix} 2x_1 + 2x_3 \\ 2x_2 + 2x_3 \\ 2x_1 + 2x_2 \end{bmatrix}$$

- 2.

$$\begin{bmatrix} 2x_2 \\ 2x_1 \\ 2x_3 \end{bmatrix}$$

- 3.

$$\begin{bmatrix} 2x_1 + 2x_2 \\ 2x_2 + 2x_3 \\ 2x_1 + 2x_3 \end{bmatrix}$$

- 4.

$$\begin{bmatrix} 2x_2 + 2x_3 \\ 2x_1 + 2x_3 \\ 2x_1 + 2x_2 \end{bmatrix}$$

Solution: Answer: option 1

Given $\mathbf{x}$ is a vector in $\mathbb{R}^{3 \times 1}$ and A is a 3×3 matrix:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Let $E = \mathbf{x}^T A \mathbf{x}$.

To find the partial derivative of E with respect to $\mathbf{x}$, we follow these steps:

1. Express E in terms of $\mathbf{x}$ and A (start from $A \times x$ then $x^T X(A \times x)$): $E = \mathbf{x}^T A \mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

2. Simplify the quadratic form: $E = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} x_1 + x_3 \\ x_2 + x_3 \\ x_1 + x_2 \end{bmatrix} E = x_1(x_1 + x_3) + x_2(x_2 + x_3) + x_3(x_1 + x_2)$

$$E = x_1^2 + x_1 x_3 + x_2^2 + x_2 x_3 + x_3 x_1 + x_3 x_2$$

Thus, the partial derivative of E with respect to $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ is: $\frac{\partial E}{\partial \mathbf{x}} = \begin{bmatrix} 2x_1 + 2x_3 \\ 2x_2 + 2x_3 \\ 2x_1 + 2x_2 \end{bmatrix}$

Other long manual way is, but compuer would prefer following:

3. Differentiate E with respect to $\mathbf{x}$: $\frac{\partial E}{\partial \mathbf{x}} = \frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^T A \mathbf{x}) = (A + A^T)\mathbf{x}$

Since A is symmetric ($A = A^T$): $\frac{\partial E}{\partial \mathbf{x}} = 2A\mathbf{x}$

4. Compute $2A\mathbf{x}$:

$$2A = 2 \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 2 & 2 \\ 2 & 2 & 0 \end{bmatrix}$$

So, the partial derivative of E with respect to $\mathbf{x}$ is: $\frac{\partial E}{\partial \mathbf{x}} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 2 & 2 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 + 2x_3 \\ 2x_2 + 2x_3 \\ 2x_1 + 2x_2 \end{bmatrix}$

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✓ Q8 Problem Statement

Let A belong to $\mathbb{R}^{M \times N}$, a rectangular matrix with Singular Value Decomposition $A = U\Sigma V^T$. Which of the following are true?

1. $UU^T = I_{M \times M}$ and $VV^T = I_{N \times N}$ ✓
2. Σ is a square matrix of eigenvalues
3. $A^T A = V\Delta V^T$ when Δ is a diagonal matrix ✓
4. $UU^T = 0_{M \times M}$ and $VV^T = I_{N \times N}$ ✓

Solution

Given the matrix $A \in \mathbb{R}^{M \times N}$ with its Singular Value Decomposition (SVD) $A = U\Sigma V^T$, let's analyze the properties of the matrices involved in the decomposition.

1. Orthogonality of U and V :

- $U \in \mathbb{R}^{M \times M}$ is an orthogonal matrix, so $UU^T = I_{M \times M}$.
- $V \in \mathbb{R}^{N \times N}$ is also an orthogonal matrix, so $VV^T = I_{N \times N}$.

2. Shape and properties of Σ :

- $\Sigma \in \mathbb{R}^{M \times N}$ is a rectangular diagonal matrix, containing the singular values of A . It is not necessarily a square matrix unless $M = N$.
- The singular values are non-negative and are placed along the diagonal of Σ .

3. Relationship between A and $A^T A$:

- $A^T A \in \mathbb{R}^{N \times N}$.
- $A^T A = (U\Sigma V^T)^T (U\Sigma V^T) = V\Sigma^T U^T U \Sigma V^T = V\Sigma^T \Sigma V^T$.
- Here, $\Sigma^T \Sigma$ is a diagonal matrix, often denoted as Δ , containing the squares of the singular values of A .
- Therefore, $A^T A = V\Delta V^T$ where Δ is a diagonal matrix.

4. Properties of U and V :

- U and V are orthogonal matrices, so $U^T U = I_{N \times N}$ and $V^T V = I_{M \times M}$.

5. Incorrect Statement:

- $UU^T = 0_{M \times M}$ is incorrect. For orthogonal matrices, $UU^T = I_{M \times M}$, not the zero matrix.

Therefore, the correct statements from your list are:

- $UU^T = I_{M \times M}$

- $VV^T = I_{N \times N}$
- $A^T A = V \Delta V^T$ where Δ is a diagonal matrix

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Q9 Problem Statement

Given a matrix A of size $D \times D$ with $\det(A) \neq 0$, find the determinant of $2A^{-1}$.

Solution

To find the determinant of $2A^{-1}$, we can use the properties of determinants.

1. The determinant of a scalar multiple of a matrix is given by: $\det(cA) = c^n \det(A)$ where c is a scalar and n is the size of the matrix. Here, $c = 2$ and $n = D$ (the dimension of the matrix A). Therefore, we have: $\det(2A^{-1}) = 2^D \det(A^{-1})$
2. The determinant of the inverse of a matrix is given by: $\det(A^{-1}) = \frac{1}{\det(A)}$
3. Since $\det(A) \neq 0$ (given), we can write: $\det(A^{-1}) = \frac{1}{\det(A)}$

Combining these results, we have: $\det(2A^{-1}) = 2^D \cdot \frac{1}{\det(A)}$

Thus, the final expression for the determinant of $2A^{-1}$ is: $\det(2A^{-1}) = \frac{2^D}{\det(A)}$

Q10 Problem Statement

Given that matrix A is invertible, evaluate the truth of the following statements:

1. A has columns that are linearly independent. ✓
2. $\|A\| = 0$.
3. $AA^{-1} = I$. ✓
4. A^2 is invertible. ✓

Solution

Let's evaluate each statement assuming A is an invertible matrix:

1. **A has columns that are linearly independent:** True. If A is invertible, its columns must be linearly independent; otherwise, the determinant would be zero, and A would not have an inverse.
2. $\|A\| = 0$: False. If A is invertible, its norm cannot be zero. The norm of a matrix is zero only if the matrix itself is the zero matrix, which is not invertible.
3. $AA^{-1} = I$: True. By definition of the inverse, multiplying a matrix by its inverse yields the identity matrix.
4. **A^2 is invertible:** True. If A is invertible, then $A^2 = A \cdot A$ is also invertible. The inverse of A^2 can be expressed as $(A^2)^{-1} = (A^{-1})^2$.